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1 Overview

The research projects I was involved in during my time at King Abdullah University of Science and Technology (KAUST) between 2009 and 2019 were in the area of enabling distributed and streaming 3D Graphics and Visualization workflows, many of them on Tiled-Display Walls, and six-sided CAVE environments at the Visualization Core Laboratory. I worked as an individual contributor between 2009 - 2013, and then served in the capacity of a Group Lead between 2013 - 2019. Our primary mission was to support domain scientists with our expertise in enabling discoveries in their specific research areas. My lab and I have contributed to the following scientific fields:

- Climate Science and Oceanography
- Combustion Physics
- Marine Ecology
- Material Science
- Computational Fluid Dynamics
- Neuroscience
- Applied Statistics

I supervised a staff of 6 scientists, and managed high-end visualization facilities in the lab (Figure 1). My contributions have been both direct and indirect. In each of the direct contributions (Sections 1.1 to 1.7), I have collaborated with domain experts directly to enable discoveries in their research. My indirect contributions (Section 1.8) are through the impact created by my staff and the use of large-scale visualization facilities in our lab. The following sections outline both.



Figure 1: Facilities at the Visualization Core Lab. Left: Tiled-Display Wall, Middle: Fully-Immersive Virtual Reality CAVE, Right: Semi-Immersive Virtual Reality Cubes.

1.1 Using Data Science to Reveal Patterns in Marine Animal movements

Research Context

Marine ecologists have been maintaining a growing database of tracked animal movements over large areas and long periods of time (eg. www.movebank.org). Specifically, distinguished KAUST marine researcher Dr. Carlos Duarte and his international collaborators have meticulously tracked movements of 321 individual Northern Elephant Seals, consisting of over 1.2 million GPS points. These animals are known to perform coordinated voyages, and communicated over 5000 Kms. There are several challenges in understanding large and complex behaviors of such animals:

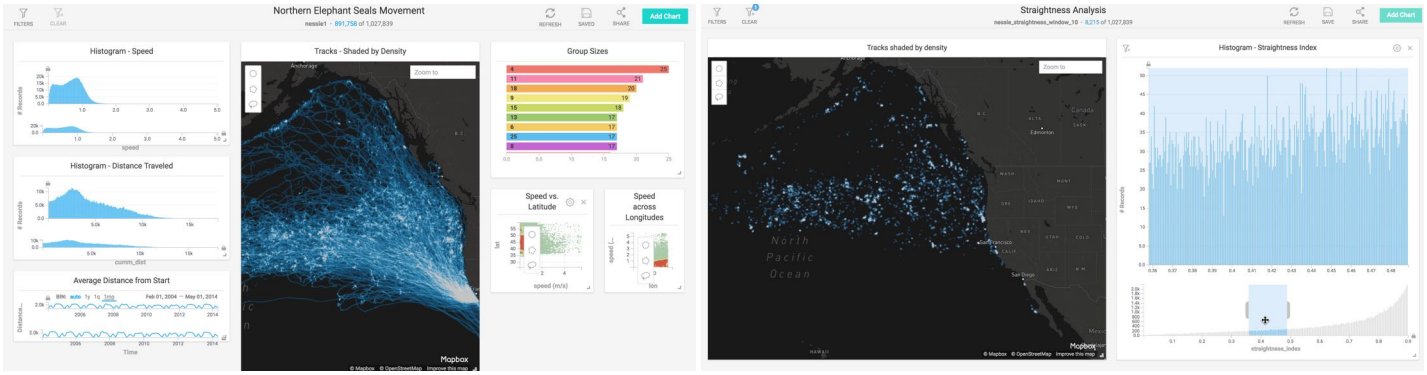


Figure 2: Left: Interactive Visualization and Analysis of Elephant Seal Tracks. Right: Discovery of foraging regions for the seals, in the Pacific Ocean.

- User-friendly querying and visualization of these GPS tracks.
- Identify emergent patterns from coordinated movements
- Identify interactions with climate and environment

Contributions and Impact

I collaborated with Dr. Duarte and Dr. Dan Costa from University of California, Santa Cruz, to:

- develop a data science algorithm to discover additional information from raw GPS tracks of 321 Northern Elephant Seals.
- develop an integrated visualization and data science tool to visualize and interactively analyze the nature of these Elephant Seal Movements (see Figure 2). The technique uses specialized hardware (GPUs) to perform these analyses quickly.
  [Click here for video]
- to help discover (through my tool) foraging regions in the Pacific Ocean, for these elephant seals. This confirms a previously published hypothesis in the field.
  [Click here for video]

Publications

I am a co-author in the following publication:

- Sonification of Animal Tracks as an Alternative Representation of Multi-Dimensional Data: A Northern Elephant Seal Example [Duarte et al., 2018]

Articles

- Translating elephant seal data into a symphony provides surprising insights
- Synchronized swimming for seal migrations

Videos

- Discovering the elephant seals foraging behavior
- Visualization and Analysis Tool for the Elephant Seal Tracks

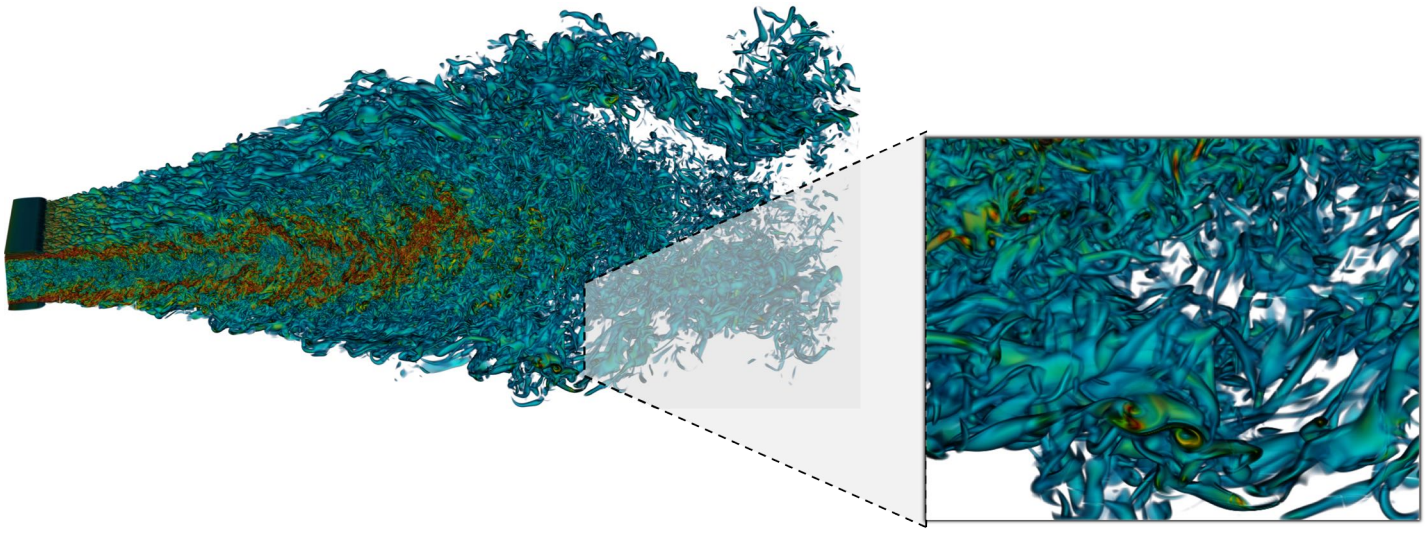


Figure 3: Visualizing turbulence structures in a 22-billion-grid point combustion simulation [Attili et al., 2016]

1.2 Visualization and Analysis of Large Scale Combustion Simulations

Research Context

Direct Numeric Simulations of Premixed Turbulent Combustion of Methane and Air models the combustion dynamics of large combustion devices such as gas turbines etc. The numerical simulation performed by KAUST researchers [Attili et al., 2016] on the KAUST Supercomputing resources calculates data for 16 different variables, over a 22-billion grid points, generating about 4.3 terabytes of data per time-step (and there are over 1000 time steps per simulation). This was the largest ever DNS simulation performed (in 2017).

These simulations feature turbulences (also called vorticities) at multiple scales (see Figure 3). Turbulence affects the quality of combustion, and therefore the efficiency of these large combustion devices. The challenge therefore is to understand the physics of these turbulence structures, and to perform an interactive, qualitative and quantitative exploration of a 4.3 terabyte dataset. This is a non-trivial task, as the data has to be distributed across a supercomputer, and the results streamed to the researcher's workstation interactively.

Contributions and Impact

With over 4 Terabytes of data per timestep, and hundreds of timesteps computed, we worked with the domain scientists to create a meaningful visualization workflow that scales with such large data sizes. Our domain scientists used several hundred cores of the Shaheen II supercomputer, a high-throughput 40-Gigabit network link between the Visualization and Supercomputing Core Labs, and a high-performance 16-node GPU cluster to produce a parallel visualization technique that can also be interactive, for large datasets. In summary, my contributions were as follows:

- Created an efficient workflow for Interactive analysis and visualization of the largest ever combustion simulation, consisting of 22-billion data points, on the KAUST Supercomputer (Shaheen II).
  [Click here for video]
- Created a novel visualization streaming workflow for on-demand exploration for combustion datasets from a supercomputer, as outlined in the paper - Query-driven parallel exploration of large datasets [Atanasov et al., 2012].
- Supervised two graduate students to implement two innovative approaches to visualize combustion datasets [Binyahib, 2013] [Nagoor, 2014] [Nagoor et al., 2015].

Publications

My contributions were used in the following publication by the domain scientist:

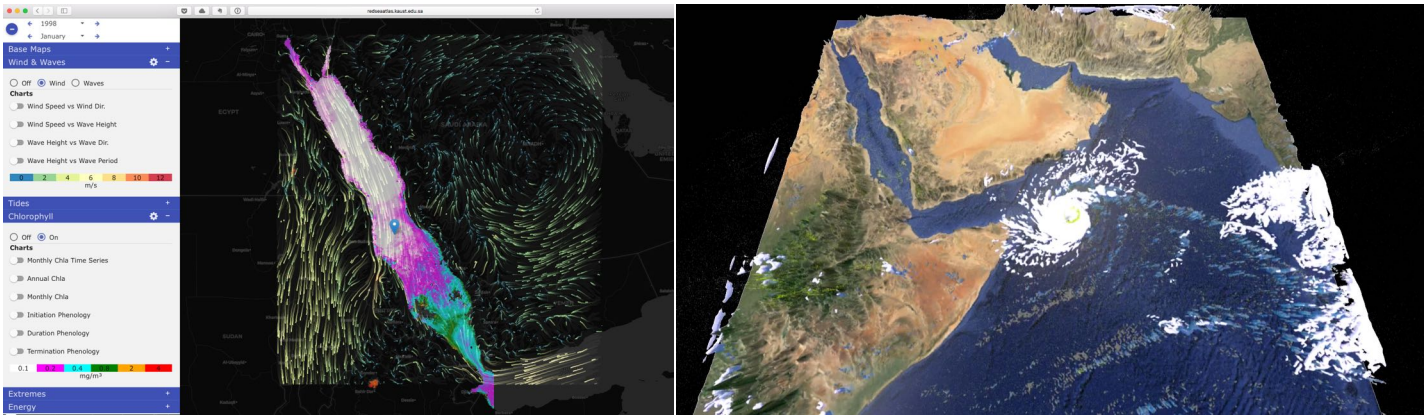


Figure 4: Left: The *RedSeaAtlas* visual-analytics tool. Right: Visualization of extreme weather events (cyclones)

- Large scale Direct Numerical Simulation of premixed turbulent jet flames at high Reynolds number [Attili et al., 2016].

I am a co-author in the following publications:

- Query-driven parallel exploration of large datasets [Atanasov et al., 2012].
- PathlinesExplorer - Image-based Exploration of Large-Scale Pathline Fields [Nagoor et al., 2015].

I supervised the following work done by M.S. students working in this research area:

- Image-Based Exploration Of Large-Scale Pathline Fields [Nagoor, 2014]
- Image-Based Exploration Of Iso-Surfaces For Large Multi-Variable Datasets Using Parameter Space [Binyahib, 2013]

Articles

- Visualizing the largest combustion simulation ever performed.

Videos

- Visualizing a Combustion Dataset
- Visualizing Streamlines (from Student Thesis Work)

1.3 Visualization and Analysis of Climate Datasets

Research Context

Climate scientists at KAUST working with ocean and atmospheric datasets deal with diverse data analysis tasks related to a variety of application areas such as tracking fisheries movements, coastal developments, environmental planning, optimization of renewable energy generation, navigational guidance for marine vessels, tourism planning, and policy-making. They encounter data from heterogeneous data sources either generated by numerical simulation models or observational data, and they often have to combine these disparate data for simultaneous analysis. These diverse data and task requirements pose great visualization challenges that are necessary to address, to enabled discoveries in climate research.

Contributions and Impact

My lab and I have been collaborating with our climate researcher, Dr. Ibrahim Hoteit, to design several custom and task-specific visual-analytics tools intended for solving problems in the ocean and atmospheric domain:

- Red Sea Atlas: We designed a custom visual analytics tool (called the *RedSeaAtlas*) intended for use by the ocean and atmospheric scientists working on different analysis tasks, dealing with multivariate ocean and atmospheric datasets for the region's Red

Sea [Afzal et al., 2019] The tool has proven to be useful to our climate researchers for:

- Tracking the flow of nutrients in the Red Sea. These nutrients are essential for marine life as they serve as a food source, and therefore identify areas most profitable for fishing.
- Analyzing extreme events in the Red Sea. These events are indicative of marine heat waves, and can cause coral bleaching if the heat-waves sustain for long periods.
- Understanding the spatiotemporal climate patterns related to wind energy in the region. These patterns are essential to optimize the locations for wind-based renewable energy infrastructure. These insights also help experts estimate extractable energy from wind at different places and then optimize the grid systems for energy generations.

[Click here for video]

- Visualizing Cyclone Chapala: The visualization in Figure 4 shows the eye of the cyclone "chapala" which formed south of the arabian peninsula in October 2015, causing significant damage to life and property in the peninsula. The cyclone was a category 4 storm with 155 mph winds, and was the second strongest tropical cyclone on record in the Arabian Sea. Over the Arabian Sea, such well structured and matured cyclones are unusual in post-monsoon periods.

  We worked with Climate Scientists (Dr. Ibrahim Hoteit) at KAUST to simulate, analyze and visualize the movements of cyclone Chapala on Shaheen II (KAUST Supercomputer). The simulation output results in several hundred gigabytes of data. We analyze and visualize these data using state of the art computational and visualization software for representing different characteristics of Chapala and to verify the accuracy of the simulation model [Theußl et al., 2016].

  [Click here for video]

Publications

I am a co-author in the following publications:

- Simulation and Visualization of the Cyclonic Storm Chapala over the Arabian Sea: A Case Study [Theußl et al., 2016]
- RedSeaAtlas: A Visual Analytics Tool for Spatiotemporal Multivariate Red Sea Data [Afzal et al., 2019]

Articles

- Seeing the Red Sea in New Ways

Videos

- RedSeaAtlas in action
- Visualizing Cyclone Chapala

1.4 KAUST-EPFL Alliance for Neuro-Inspired High Performance Computing

Research Context

The KAUST-EPFL Alliance for Neuro-inspired high performance computing was established in 2013 between Prof. Henry Markram's Blue Brain Lab at EPFL - École polytechnique fédérale de Lausanne, KAUST's Dean of Biological Sciences and Environment division - Dr. Pierre Magistretti, and my lab. The goal of this alliance was to develop biologically detailed 3D models of the blood supply to the brain, and study the glial cells coupling neurons and vasculature in the neuro-vascular-glial unit. These detailed models of the brain will facilitate understanding on how to perform computations with minimal energy requirements and with high reliability, on future neuro-inspired supercomputing systems. This will enable simulation, visualization and analysis to be performed at the same time on the same machine, interactively, laying the foundation for exploiting supercomputers for science and engineering in novel ways.

Contributions and Impact

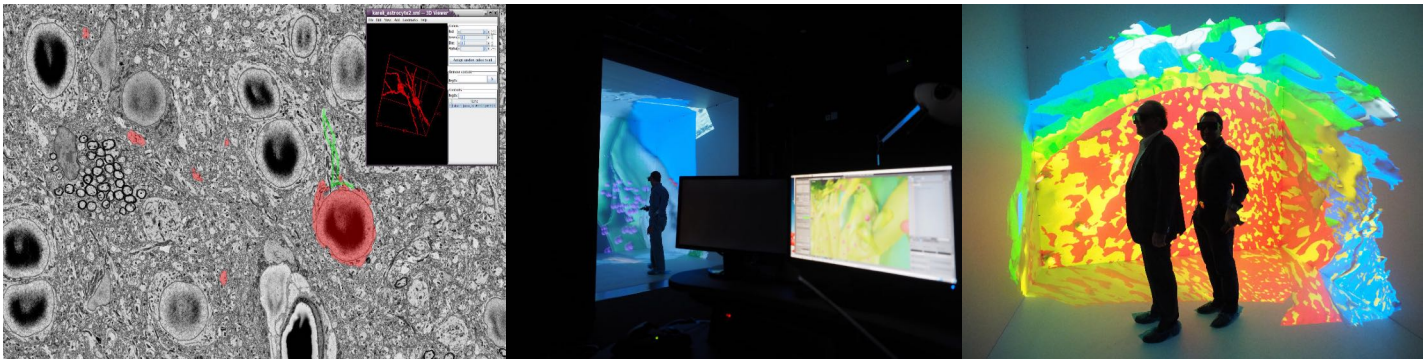


Figure 5: Reconstructing brain neurons (left) and visualizing them in my lab's CORNEA Virtual Reality Environment (center, and right)

- As a collaborating partner, my lab and I worked with scientists at KAUST and EPFL to provide visualization expertise and workflows, and reference configurations for both hardware and software for EPFL's own visualization facilities dedicated for the project. A letter of invite, as part of the collaboration with EPFL is provided here
- I hired the founding team that would provide the Visualization and 3D reconstruction expertise for the project at KAUST. The team was trained and incubated in my lab, before branching out and working full-time with Prof. Pierre Magistretti.
- Providing innovations in high-resolution 3D Reconstruction: For reconstructing 3D structures of neurons (see Figure 5), cross-sections of mouse brain tissue scanned under an electron microscope were reconstructed using special software, allowing a multidisciplinary team of researchers to examine the tissue in a 3D virtual environment in the lab. We created innovative improvements in 3D Reconstruction techniques to handle large imaging stacks (26GB). Such a stack would have required 1TB RAM and several weeks to months, to accurately reconstruct all features. My lab created workflows to perform 3D reconstruction on large image stacks in just a few hours, on a single workstation [Holst et al., 2016].
- Using the lab's high-end virtual-reality facilities: Due to the complexity of the reconstructed 3D models of brain tissues, qualitative observations are difficult even for an experienced neuroscientist. With the help of visualization and virtual reality facilities and software we developed in my lab, researchers could "walk" into a 3D model containing neuronal and astrocytic processes. Many qualitative and quantitative discoveries were enabled through this workflow [Calì et al., 2016] [Calì et al., 2017] [Agus et al., 2018]. For instance, the use of CORNEA Virtual Reality in the lab was key to the observation of a non-random distribution of glycogen, and led the team to develop tools to quantitatively analyze glycogen clustering and proximity to other sub-cellular features [Calì et al., 2016].

Publications

I am a co-author in the following publication:

- Three-dimensional immersive virtual reality for studying cellular compartments in 3D models from EM preparations of neural tissues [Calì et al., 2016]

My lab's resources have been used and acknowledged in the following publications:

- Adding large EM stack support [Holst et al., 2016]
- Visual Analysis of Glycogen Derived Lactate Absorption in Dense and Sparse Surface Reconstructions of Rodent Brain Structures [Calì et al., 2017]
- GLAM: Glycogen-derived Lactate Absorption Map for visual analysis of dense and sparse surface reconstructions of rodent brain structures on desktop systems and virtual environments [Agus et al., 2018]

Articles



Figure 6: Identifying and Visualizing new turbulence patterns in particle-driven gravity currents, on the lab's 40-megapixel tiled-display wall

- The Brain Gamers
- Exploring the Brain by Mimicking the Brain

Videos

- 3D Reconstruction of Hippocampus for visual analysis of brain structures
- Use of CORNEA Virtual Reality Environment for 3D Reconstructed Glia and Neuron Cells
- A scientific narrative produced by my lab, in collaboration with Dr. Magistretti's group

1.5 Visualizing Turbulence Patterns in Particle Driven Gravity Currents

Research Context

Particle driven currents simulate and study the flow of silt-filled water (i.e. rivers carrying sand, clay) into highly saline water bodies (oceans, seas). Such flows deposit large amounts of silt, clay and other sediments on the ocean floor over several years, and determine the topography and geology of the floor.

Contributions and Impact

I worked with the domain scientists to visualize new turbulence patterns that were discovered for the first time. Specifically, two interacting "instabilities" were discovered - Kelvin-Helmholtz instability (due to shear) and "fingering" instability (due to gravity). These interactions are quite complex and hard to observe experimentally. The instabilities were visualized and observed for the first-time ever through this work (see Figure 6).

These patterns determine how sediments are deposited on the sea-floor. Therefore, modeling, visualizing and analyzing this phenomena is critical for many reasons - for determining the laying of sub-marine communication cable pathways, for evaluating viable drilling regions of off-shore hydrocarbon deposits etc. For instance, an off-shore drilling operation can cost up to $200 million versus $10 million for on-shore drilling.

Publications

My lab's resources have been used and acknowledged in the following publications:

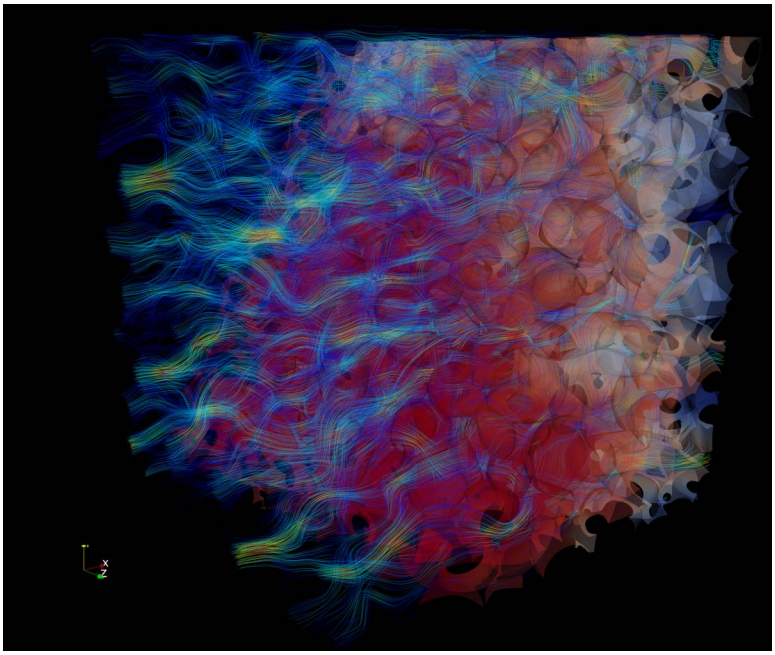


Figure 7: Observing flow-characteristics through a porous media.

- An Analysis of Mesopicnal Flows: Intrusive Density- and Particle-Driven Gravity Currents via Direct Numerical Simulations [Espath et al., 2019]

Videos

- Dynamics of turbulent structures in particle-driven gravity currents

1.6 Characterizing Fluid-Flow Through Porous Media

Research Context

Fluid flow through porous media is the manner in which fluids behave when flowing through a porous medium, for example sponge or wood, or when filtering water using sand or other porous material. The phenomena is also observed in geophysics, when extracting hydrocarbons from reservoirs, where oil and gas flows pores inside rocks.

Contributions and Impact

A detailed visualization that shows the phenomena of "flow" through a porous rock sample was implemented (see Figure 7). The geometry for the rock sample was obtained through Micro-CT - a technique that allows a non-destructive 3D imaging and reconstruction of objects. The streamlines show the flow of a fluid through the void-spaces in the sample, with colors depicting the speed of the flow (red is high velocity, blue is low velocity). Such visualization and analysis of flow models can be used to understand multi-phase flow (eg. oil, water) through rocks underground, for instance.

Publications

My lab's resources have been used and acknowledged in the following publications:

- Pore-scale simulation of fluid flow and solute dispersion in three-dimensional porous media [Icardi et al., 2014]

1.7 Other Contributions

The following list summarizes my contribution in other specific areas, as indicated:

- Implementation of hardware and software workflows for Virtual Reality facilities, and using them for training non-experts:

  - The future of the CAVE [DeFanti et al., 2011].
  - Acquisition of stereo panoramas for display in VR environments [Ainsworth et al., 2011].
  - Towards a digital makkah – Using immersive 3d environments to train and prepare pilgrims [Schneider et al., 2011].
- 3D Animation and Behavior modeling of 3D virtual characters, using Computer Graphics and Machine Learning:
  - Semi-automatic End-user Tools for Construction of Virtual Avatar Behaviors [**Srinivasan, M.** and Metoyer, 2011]
  - A Data-Driven Model of Pedestrian Movement [Casburn et al., 2007]
  - Controllable Real-time Locomotion Using Mobility Maps [**Srinivasan, M.** et al., 2005]
  - A responsiveness metric for controllable characters [**Srinivasan, M.** and Metoyer, 2005]
  - A Tangible Interface for High-Level Direction of Multiple Animated Characters [Metoyer et al., 2003]

1.8 Indirect Contributions

The following contributions were enabled by staff and facilities from my lab. I have supervised, consulted with scientists, and provided inputs to each of these activities.

- Improving Pore Network Imaging & Characterization of Microporous Carbonate Rocks Using Multi-Scale Imaging Techniques: We supported 3D reconstruction of rock-core samples that were CT-Scanned. The reconstruction helps quantify pore characteristics in rock samples excavated from potential oil-drilling sites, which in turn allows the estimation of oil-reserves in the area [Hassan et al., 2017].
- Experimental and numerical characterization of the water flow in spacer-filled channels of spiral-wound membranes: We helped the researchers at KAUST's Water Desalination Research Center to visualize the flow characteristics of water over de-salination membranes. Our contributions help them understand how de-salination membranes degrade and fail, and to design efficient membranes that are cost-effective [Bucs et al., 2015].
- Visualization for Spatio-temporal Statistics: The latest advances in computing and virtual reality (VR) in my lab have enabled statistics researchers at KAUST to develop a suite of apps that allow users to visualize and interpret large and complex datasets in three dimensions.
  - Paper: Visuanimation in Statistics [Genton et al., 2015]
  - Paper: Visualizing spatiotemporal models with virtual reality: from fully immersive environments to applications in stereoscopic view [Castruccio et al., 2019]
  - Article: New stats apps show a virtual reality
- Data Science training program: With an increase in demand for the use of data science tools and workflows in scientific areas at KAUST, I provided a strategic direction for my lab to hire a Data Scientist (2018) and institute a strong data-science program for training and research.
  - Article: Introduction to Data Science Workshops - Learning the tools of the trade.

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